

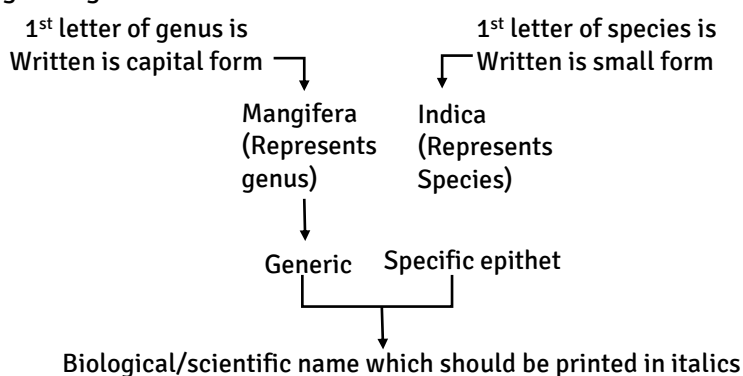
Exercise 1.1

1. Metabolism is exhibited by :
 - (1) All plants
 - (2) All animals
 - (3) Fungi & Microbes
 - (4) All of these
2. Which of the following is **not** the defining property of living ?
 - (1) Consciousness
 - (2) Reproduction
 - (3) Growth (intrinsic)
 - (4) Metabolism
3. In higher plants & animals, growth & reproduction are linked but are
 - (1) Mutually exclusive events
 - (2) Mutually inclusive events
 - (3) Mutually inclusive events along with monerans
 - (4) Mutually exclusive events along with protistans
4. Which of the following is not applicable to living organisms?
 - (1) These are self replicating and evolving system
 - (2) These are self replicating and capable of responding to external stimuli
 - (3) These are self replicating and non-regulating interactive system
 - (4) They exhibit metabolism
5. Most obvious and technically complicated feature of all living organisms is
 - (1) Metabolism
 - (2) Cellular organisation
 - (3) Growth
 - (4) Consciousness
6. Number of known species of Earth is
 - (1) 1.6 to 2.7 billions
 - (2) 1.7 to 1.8 millions
 - (3) 1.6 to 1.7 thousands
 - (4) 1.6 to 1.7 crores
7. Which of the following statements regarding the response of living organisms to external stimuli is **correct** ?
 - (1) The external environmental stimuli can be physical and chemical only.
 - (2) All organisms, from the prokaryotes to the most complex eukaryotes can sense and respond to environmental stimuli.
 - (3) Consciousness and response to external stimuli is the defining property of living organisms.
 - (4) Both (2) and (3)
8. Which of the following is a defining characteristic of living organisms?
 - (1) Growth
 - (2) Ability of make sound
 - (3) Reproduction
 - (4) Response to external stimuli
9. The living organisms can be unexceptionally distinguished from the non-living things on the basis of their ability for :-
 - (1) Photosynthesis
 - (2) Metabolism
 - (3) Growth and movement
 - (4) Reproduction
10. All living organisms are linked to one another because :-
 - (1) They have common genetic material of the same type
 - (2) They share common genetic material but to varying degrees
 - (3) All have common cellular organization
 - (4) All of above

- (2) First letter of generic name should be in capital letter and first letter of specific epithet should be in small letter. **E.g. *Mangifera indica*.**
- (3) When written with free hand or typed, then generic name and specific epithet should be underlined separately. But during printing name should be in italics to indicate their latin origin.
- (4) Name of the scientist who proposed nomenclature should be written in short after the specific epithet **E.g. *Mangifera indica* Linn.**
- (5) Name of scientist should be neither underlined nor in italics but written in Roman letters (simple alphabets)
- (6) Scientific names should be derived from Latin (usually) or Greek languages because they are dead languages.
- (7) In plant nomenclature tautonyms are not valid i.e. generic name and specific epithet should not be same in plants **E.g. *Mangifera mangifera*** is not correct.

But tautonyms are valid in animal nomenclature (ICZN-International Code of Zoological Nomenclature) **E.g. *Naja naja*** (Indian cobra), ***Rattus rattus*** (Rat).

Method of writing biological or scientific name can be memorised as



Exercise 1.2

1. In Binomial Nomenclature,
 - (1) Genus name is written after species
 - (2) Genus and species names are written in italics.
 - (3) Genus and species have the same name
 - (4) The first letter of genus and species name is capital.
2. The standard name of any living organism is known by the same name all over the world is called :
 - (1) Nomenclature (2) Taxonomy
 - (3) Classification (4) Description
3. The scientific naming system of two words was first used by
 - (1) Aristotle
 - (2) C. Linnaeus
 - (3) Bentham and Hooker
 - (4) Theophrastus
4. The process by which any thing is grouped into convenient categories based on some easily observable characters is called :
 - (1) Identification (2) Systematics
 - (3) Classification (4) Nomenclature
5. Systematics was first time proposed by
 - (1) C. Linnaeus in Pinax
 - (2) Casper Bauhin in Pinax
 - (3) Casper Bauhin in Species Plantarum
 - (4) C. Linnaeus in *Systema Naturae*
6. Who wrote systema naturae:-
 - (1) Linnaeus (2) Mayr
 - (3) John Ray (4) De Candolle
7. A taxon is:
 - (1) A group of related families
 - (2) A group of related species
 - (3) A type of living organisms
 - (4) A taxonomic group of any ranking

8. The process in which biologists follow universally accepted principles to provide name of any organism is called as:
(1) Identification
(2) Classification
(3) Nomenclature
(4) Systematics
9. Most of the scientific names are drawn from the following language -
(1) German
(2) Greek
(3) Latin
(4) Spanish
10. Taxonomy refers to :-
(1) Plant classification
(2) Plant nomenclature
(3) Plant affinity
(4) All the above
11. ICBN was first published in :
(1) 1961 (2) 1964
(3) 1975 (4) 1753
12. What are the essential basis of modern taxonomic studies?
(1) External and internal structure
(2) Structure of cell
(3) Developmental process and ecological information
(4) All of these
13. As we go from species to kingdom in a taxonomic hierarchy, the number of common characteristics-
(1) Will decrease
(2) Will increase
(3) Remain same
(4) May increase or decrease
14. The number and types of organisms present on earth make:
(1) Taxonomy
(2) Plant diversity
(3) Animal diversity
(4) Biodiversity
15. First act in taxonomy is :-
(1) Characterisation
(2) Identification
(3) Naming
(4) Classification
16. Which of the following statements regarding nomenclature is correct :-
(1) Generic name always begins with capital letter whereas specific name with small letter
(2) Scientific name should be printed in italics
(3) Scientific name when typed or handwritten should be underlined
(4) All the above
17. Phylogeny refers to :-
(1) Natural classification
(2) Evolutionary classification
(3) Evolutionary history
(4) Origin of algae
18. Artificial system of classification classifies plants on the basis of -
(1) One or two characters
(2) Phylogenetic trends
(3) Many naturally existing characters
(4) Gross Morphological Characters
19. Systematics deals with :-
(1) Classification
(2) Nomenclature
(3) Plant description
(4) Identification
20. The term 'Systematics' refers to :-
(1) Identification and classification of plants and animals
(2) Nomenclature and identification of plants and animals
(3) Diversity of kinds of organisms and their relationship
(4) Different kinds of organisms and their classification

Organisms with their Taxonomic Categories						
Common Name	Biological Name	Genus	Family	Order	Class	Phylum/ Division
Man	<i>Homo sapiens</i>	<i>Homo</i>	Hominidae	Primata	Mammalia	Chordata
Housefly	<i>Musca domestica</i>	<i>Musca</i>	Muscidae	Diptera	Insecta	Arthropoda
Mango	<i>Mangifera indica</i>	<i>Mangifera</i>	Anacardiaceae	Sapindales	Dicotyledonae	Angiospermae
Wheat	<i>Triticum aestivum</i>	<i>Triticum</i>	Poaceae	Poales	Monocotyledonae	Angiospermae

Note:– As we go higher from species to kingdom number of common characters decreases. Lower the taxa more are the characteristics that the members with in the taxon share.

Higher the category, greater is the difficulty of determining the relationship to other taxa at the same level.

Exercise 1.3

- Which of the following have maximum common characters ?
 (1) Felidae, Canidae
 (2) Potato, Brinjal
 (3) Carnivora, Primata
 (4) *Datura*, *Petunia*
- Which of the following statement is **correct** ?
 (1) Potato and brinjal are two different genera
 (2) Lion and tiger are placed in different genera
 (3) *Petunia* and *Datura* are placed in same family
 (4) Solanaceae and Poaceae are placed in same class.
- The process of classification of all living organism into different taxa based on their characteristics is referred as :
 (1) Taxonomy
 (2) Systematics
 (3) Nomenclature
 (4) Identification
- Select the **correct** option w.r.t. hierarchial arrangement of taxonomic categories in ascending order.
 (1) Division → Class → Order → Family
 (2) Genus → Family → Order → Class
 (3) Species → Order → Class → Family
 (4) Kingdom → Order → Family → Genus
- Taxonomic category with a group of related genera is characterized on the basis of :
 (1) Cell structure
 (2) Vegetative features
 (3) Only floral characters
 (4) Vegetative and reproductive features
- In ascending order of taxon common morphological characters :
 (1) Decrease
 (2) Increase
 (3) First decrease than increase
 (4) First increase than decrease
- Taxonomic categories are __A__ biological entities & __B__ morphological aggregates. A & B respectively stand for :
 (1) Same, Merely
 (2) Distinct, Not merely
 (3) Same, Not merely
 (4) Distinct, Merely

Species Concept :

- **John Ray** : Proposed the term and concept of species.

Biological Concept of Species :

- (1) **Ernst Mayr** (Darwin of 20th century) proposed the **biological concept of species**.
- (2) According to Mayr "Species is the group of individual which can interbreed naturally and produce fertile offspring. This definition of Mayr was incomplete because this definition is applicable to only on sexually reproducing living beings but there are many organisms that have only asexual mode of reproduction.

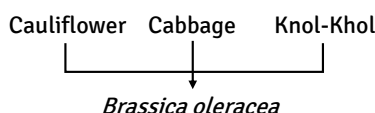
E.g. Bacteria, Mycoplasma. BGA

- (3) Biological species concept is based on **reproductive isolation**.

In taxonomy, the determination of species is mainly based on morphological characters

Taxonomic concept of species :

- When the species is determined on the basis of morphological characters then it is called as taxonomic species. eg. If two plants have almost same fundamental morphological characters, then they belong to same species.

Knowledge Booster :

These three have same morphological characters. Therefore they belong to same taxonomic species but they are not interbreed naturally so they belong to three different biological species.

In higher plants the determination of species is mainly based on the morphology of flower (floral morphology) because floral (reproductive) characters are more conservative as compared to vegetative.

Exercise 1.4

- | | |
|---|--|
| <ol style="list-style-type: none"> 1. The basic smallest unit of classifications is :-
 (1) Genus (2) Species
 (3) Order (4) All of the above 2. What are the basis of modern classification for taxonomists :-
 (1) External and internal structure of organism
 (2) Development process of organism
 (3) Ecological informations of organism
 (4) All of these 3. Plant nomenclature means :-
 (1) To give names to plants without any rules
 (2) Nomenclature of plants under the international rules
 (3) Nomenclature of plants in local language
 (4) Nomenclature of plants in English language | <ol style="list-style-type: none"> 4. Descending arrangement of categories is called :-
 (1) Classification (2) Taxonomy
 (3) Hierarchy (4) key 5. For higher plants, flowers are chiefly used as a basis of classification, because-
 (1) These show a great variety in colour
 (2) It can be preserved easily
 (3) Reproductive parts are more conservative than vegetative parts
 (4) Reproductive parts are less conservative 6. Who is regarded as "Darwin of 20th century" ?
 (1) John Ray (2) Lamarck
 (3) Ernst Mayr (4) Darwin |
|---|--|

7. Group of organisms that closely resemble each other and freely interbreed in nature, constitute a:-
 (1) Species (2) Genus
 (3) Family (4) Taxon
8. Which arrangement is correct ascending order?
 (1) Species < genus < order < family
 (2) Genus < species < family < order
 (3) Order < family < Genus < species
 (4) Species < genus < family < order
9. Nomenclature is very must in taxonomy which of the following is **not correct** about scientific nomenclature ?
 (1) They ensure that each organism has only one name
 (2) They also ensure that such a name has not been used for any other known organism
 (3) Scientific nomenclature is a standardised naming system
 (4) Different countries of the world use different kinds of scientific nomenclature system
10. Species are considered as :-
 (1) Real basic units of classification
 (2) The lowest units of classification
 (3) Artificial concept of human mind
 (4) Biggest units of classification devised by taxonomists
11. Mayr proposed which type of concept of species :-
 (1) Static concept
 (2) Biological concept
 (3) Topological concept
 (4) Genetic concept
12. A division is formed by combining several -
 (1) Orders
 (2) Families
 (3) Classes
 (4) Tribes
13. All the categories used in classification of organisms constitute :-
 (1) Taxonomy
 (2) Systematics
 (3) Taxonomic hierarchy
 (4) Taxonomic affinity
14. In potato, brinjal, lion, leopard, and tiger. how many species and genera are there respectively ?
 (1) 4 & 2 (2) 5 & 3
 (3) 5 & 4 (4) 5 & 2
15. The correct sequence of plant taxonomic categories is :-
 (1) Phylum-order-class-family-genus-species
 (2) Division-class-order-family-genus-species
 (3) Division-class-family-order-genus-species
 (4) 1 and 2 both
16. For declaration of new species of higher plants what characters are used :-
 (1) Floral character of new species
 (2) Anatomical character of new species
 (3) Physiological character of new species
 (4) Character of endosperm
17. An organisms is in the same class but not in the same family. It may belong to same :-
 (1) Genus (2) Species
 (3) Variety (4) Order
18. In the names *Mangifera indica*, *Solanum tuberosum* and *Panthera leo*, the words *indica*, *tuberosum* and *leo* represent ?
 (1) Generic name (2) Generic epithet
 (3) Specific name (4) Specific epithet
19. Which of the following is a correct name :-
 (1) *Solanum tuberosum*
 (2) *Solanum Tuberosum*
 (3) *Solanum tuberosum* Linn.
 (4) All the above
20. The relation of solanaceae and convolvulaceae with polymoniales is similar to the relation occurring in :
 (1) Felidae and Canidae with carnivora
 (2) Primata and carnivora with Mammalia
 (3) Amphibia and Reptilia with Chordata
 (4) *Solanum* and *Petunia* with solanaceae
21. Animals, mammals and dogs represent :-
 (1) Taxa at different levels
 (2) Taxa at same level
 (3) Different levels of same taxa
 (4) All are correct

Exercise 2

1. Why the determination of relationship becomes more complex in higher taxonomic categories?
- (1) Number of common characters goes on decreasing in lower taxa.
 - (2) Number of common characters goes on decreasing in higher taxa.
 - (3) Because classification itself is very difficult process.
 - (4) Number of common characters goes on increasing in higher taxa
2. Which of the following is not a **correct** statement?
- (1) Order is the assemblage of families which exhibit a few similar characters in comparison to families.
 - (2) Convolvulaceae and Solanaceae families are included in order polymoniales mainly on the basis of floral characters.
 - (3) In hierarchy, both broad categories and sub categories are used.
 - (4) Class is a sub category which includes related order.
3. Which of the following is not true for a species?
- (1) Variations occur among members of a species
 - (2) Gene flow does not occur between the populations of a species
 - (3) Each species is reproductively isolated from every other species
 - (4) Members of a species can interbreed
4. Study the following statements carefully and give the answer :
- A. Biological names are generally in Latin and printed in italics.
 - B. The first word in biological names represents the genus.
 - C. Both the words in a biological name are separately underlined when printed.
 - D. The first word of biological name starts with capital letter when given in memory of a person otherwise with small letter.
- (1) A, D - correct B, C-correct
 - (2) A, C - correct B, D - incorrect
 - (3) A, B - correct C, D - incorrect
 - (4) A, B - incorrect C, D - correct
5. Which of the following is / are **not** correctly match ?
- | | Common Name | Genus | Family |
|-------|-------------|-----------|------------|
| (i) | Man | Homo | Hominidae |
| (ii) | Housefly | Musca | Muscidae |
| (iii) | Mango | Mangifera | Sapindales |
| (iv) | Wheat | Triticum | Poaceae |
- (1) i & ii (2) only i
 - (3) only iii (4) iii & iv
6. How many are the incorrect statements from the followings?
- (A) Lower the taxa, more are the characteristics that the members within the taxon share
 - (B) Higher the taxonomic category, greater is the difficulty of determining the relationship to other taxa at the same level
 - (C) In lower taxa the problems of classification becomes more complex
 - (D) As we go higher from species to kingdom, the number of common characters increases
- (1) One (2) Two
 - (3) Three (4) Four

7. Which of the following statements about classification is not true
- Members of a family are less similar than members of an included genus
 - An order has more members than the number of members in an included genus
 - Families have more members than phyla
 - The number of species in a taxon depends on their relative degree of similarity.

- (1) Only III (2) Only IV
(3) Only II (4) Only I

8. Characters drawn from flowers are mainly used as basis of classification in seed plants because :-

- Flowers show great variation in colour and traits
- Flowers are preserved easily
- Flowers are easy and nice to work
- Reproductive parts are more conservative than vegetative parts and show less variation

9. Read the following statements:

- Classification is a single step process
- Classification involves hierarchy of steps in which each step represents a rank or category
- Taxon is a unit of ecological hierarchy
- Insects represents a species
- The taxonomic group/categories are distinct biological entities and not merely morphological aggregates

How many of the above statements are **correct**?

- (1) Five (2) Three
(3) Two (4) One

10. Two plants can be conclusively said to belong to the same species if they :

- Have same number of chromosomes
- Can reproduce freely with each other and form seeds
- Have more than 90 percent similar genes
- Looks similar and possess identical secondary metabolites

11. Consider the following tabel and choose the correct sequence of A, B, C, D, E, F, G and H :-

Common name	Biological Name	Genus	Family	Order	Class
Man	Homo sapiens	Homo	A	Primata	B
Housefly	Musca Domestica	Musca	C	D	Insecta
Mango	Mangifera indica	Mangifera	E	F	Dicotyle donae
Wheat	Triticum asetivum	Triticum	Poaceae	G	H

- A-Muscidae, B-Mammalia, C-Hominidae, D-Diptera, E-Anacardiaceae, F-Sapindales, G-Poales, H- Monocotyledonae
- A-Hominidae, B-Muscidae, C- Mammalia, D-Diptera, E-Anacardiaceae, F-Sapindales, G-Poales, H-Dicotyledonae
- A-Hominidae, B-Mammalia, C-Muscidae, D-Diptera, E-Anacardiaceae, F-Sapindales, G-Poales, H-Monocotyledonae
- A-Hominidae, B-Mammalia, C-Muscidae, D-Anacardiaceae, E- Diptera, F-Poales, G-Sapindales, H-Monocotyledonae

12. Match the followings :-

(a)	Yeast	(i)	Regeneration
(b)	Hydra	(ii)	Budding
(c)	Protonema of mosses	(iii)	Fragmentation
(d)	Filamentous algae		
(e)	Planaria		
(f)	Flat worm		

- a-ii, b-i, c-ii, d-ii, e-i, f-iii
- a-iii, b-iii, c-ii, d-i, e-i, f-iii
- a-iii, b-i, c-iii, d-ii, e-ii, f-i
- a-ii, b-ii, c-iii, d-iii, e-i, f-i

13. Select the correct combination :

- Plant : limited cell division.
 - Animals : Cell division throughout life span in all types of cells.
 - Unicellular organisms : Growth observation by microscopes
 - Living organism : External growth
- (1) a (2) b
(3) c (4) d

14. Isolated metabolic reactions in-vitro are:
 (1) Neither living nor non-living things
 (2) Non-living things
 (3) Living reactions
 (4) Always faulty reactions
15. Which one of the following animals is correctly matched with its particular named taxonomic category?
 (1) Tiger - Tigris, the species
 (2) Cuttlefish - Mollusca, a class
 (3) Humans - Primata, the family
 (4) Housefly - Musca, an order
16. In which of the following taxon, other three are included itself?
 (1) Phylum
 (2) Order
 (3) Class
 (4) Family
17. 'X' and 'Y' are the components of binomial nomenclature. This naming system was proposed by 'Z'?
 (1) X-Generic name, Y-Specific epithet, Z-Carolus Linnaeus
 (2) X-Specific epithet, Y-Generic name, Z-R.H. Whittaker
 (3) X-Specific epithet, Y-Generic name, Z-Carolus Linnaeus
 (4) X-Generic name, Y-Specific epithet, Z-R.H. Whittaker

18. Match the following and choose the correct option.

Column - I		Column - II	
(a)	Family	(i)	Tuberosum
(b)	Kingdom	(ii)	Polymoniales
(c)	Order	(iii)	Solanum
(d)	Species	(iv)	Plantae
(e)	Genus	(v)	Solanaceae

- (1) a-iv, b-iii, c-v, d-ii, e-i
 (2) a-v, b-iv, c-ii, d-i, e-iii
 (3) a-iv, b-v, c-ii, d-i, e-iii
 (4) a-iv, b-iii, c-ii, d-i, e-v

19. Match the following :

	Column- I		Column-II
(a)	Kingdom	(i)	First name of the organism
(b)	Division	(ii)	Second form of scientific name
(c)	Class	(iii)	Plantae
(d)	Order	(iv)	Ae
(e)	Family	(v)	Ales
(f)	Genus	(vi)	Phyta
(g)	Species	(vii)	aceae

- (1) a-iv, b-vii, c-vi, d-v, e-i, f-iii, g-ii
 (2) a-ii, b-i, c-iii, d-iv, e-v, f-vii, g-vi
 (3) a-i, b-ii, c-iv, d-iii, e-vi, f-vii, g-v
 (4) a-iii, b-vi, c-iv, d-v, e-vii, f-i, g-ii

20. Match the columns :-

	Column- I		Column-II
(a)	ICVN	(i)	International code of botanical Nomenclature
(b)	ICBN	(ii)	International code of bacteria Nomenclature
(c)	ICZN	(iii)	International code of virus Nomenclature
(d)	ICNB	(iv)	International code of zoological Nomenclature

- (1) a-i, b-ii, c-iii, d-iv
 (2) a-iii, b-i, c-iv, d-ii
 (3) a-iv, b-iii, c-ii, d-i
 (4) a-iv, b-ii, c-i, d-iii

Exercise 3

1. **Assertion :-** Binomial names are latinised or derived from Latin irrespective of their origin.
Reason :- Latin is a dead language, so used by scholars.
(1) If both assertion and reason are true and reason is the correct explanation of assertion.
(2) If both assertion and reason are true but reason is not the correct explanation of assertion.
(3) If assertion is true but reason is false.
(4) If both assertion and reason are false
2. **Assertion:-** Modern taxonomic studies provide more accurate grouping of various organisms.
Reason:- Modern taxonomic studies are based on external and internal structures along with the structure of cell development process and ecological informations.
(1) If both assertion and reason are true and reason is the correct explanation of assertion.
(2) If both assertion and reason are true but reason is not the correct explanation of assertion.
(3) If assertion is true but reason is false.
(4) If both assertion and reason are false.
3. **Assertion:-** Family has a group of related orders and classes.
Reason:- Families are characterised on the basis of only vegetative characters not by reproductive characters.
(1) If both assertion and reason are true and reason is the correct explanation of assertion.
(2) If both assertion and reason are true but reason is not the correct explanation of assertion.
(3) If assertion is true but reason is false.
(4) If both assertion and reason are false.
4. **Assertion:-** Genera are aggregates of closely related species.
Reason:- Genus comprises a group of related species which has more characters in common in comparison to species of other genera.
(1) If both assertion and reason are true and reason is the correct explanation of assertion.
(2) If both assertion and reason are true but reason is not the correct explanation of assertion.
(3) If assertion is true but reason is false.
(4) If both assertion and reason are false.
5. **Assertion:-** In higher category there is great difficulty of determining the relationship to other taxa at the same level.
Reason:- In higher taxa there are less characteristics that the members within the taxa share
(1) If both assertion and reason are true and reason is the correct explanation of assertion.
(2) If both assertion and reason are true but reason is not the correct explanation of assertion.
(3) If assertion is true but reason is false.
(4) If both assertion and reason are false.
6. **Assertion:-** Biological concept of species is based on reproductive isolation.
Reason:- Biological concept of species is universal
(1) If both assertion and reason are true and reason is the correct explanation of assertion.
(2) If both assertion and reason are true but reason is not the correct explanation of assertion
(3) If assertion is true but reason is false.
(4) If both assertion and reason are false.

7. Assertion:- Classification of Linnaeus is artificial.

Reason:- It is based on one or two characters.

- (1) If both assertion and reason are true and reason is the correct explanation of assertion.
- (2) If both assertion and reason are true but reason is not the correct explanation of assertion.
- (3) If assertion is true but reason is false.
- (4) If both assertion and reason are false.

8. Assertion (A): For plants scientific names are based on agreed principles and criteria which are provided in ICNB.

Reason (R): The scientific names ensure that each organism has only one name, but this name can be repeated for other organism of same kingdom.

- (1) Both (A) & (R) are true and the (R) is the correct explanation of the (A)
- (2) Both (A) & (R) are true but the (R) is not the correct explanation of the (A)
- (3) (A) is true but (R) is false
- (4) Both (A) and (R) are false

9. Assertion (A): Binomial naming system given by Carolus Linnaeus is being practiced by biologists all over the world.

Reason (R): This naming system using a two word format was found convenient.

- (1) Both (A) & (R) are true and the (R) is the correct explanation of the (A)
- (2) Both (A) & (R) are true but the (R) is not the correct explanation of the (A)
- (3) (A) is true but (R) is false
- (4) Both (A) and (R) are false

10. Assertion (A): Species is the most fundamental unit of taxonomy.

Reason (R): In taxonomic hierarchy, species shows minimum common features.

- (1) Both (A) & (R) are true and the (R) is the correct explanation of the (A)
- (2) Both (A) & (R) are true but the (R) is not the correct explanation of the (A)
- (3) (A) is true but (R) is false
- (4) Both (A) and (R) are false

11. Assertion (A): Families are characterized on the basis of both vegetative and reproductive features of species.

Reason (R): Members of family can show morphological similarities, but unexceptionally none of the genera of same family can interbreed with each other.

- (1) Both (A) & (R) are true and the (R) is the correct explanation of the (A)
- (2) Both (A) & (R) are true but the (R) is not the correct explanation of the (A)
- (3) (A) is true but (R) is false
- (4) Both (A) and (R) are false

12. Assertion (A): Classification of Linnaeus is artificial.

Reason (R): It is based on artificial characters.

- (1) Both (A) & (R) are true and the (R) is the correct explanation of the (A)
- (2) Both (A) & (R) are true but the (R) is not the correct explanation of the (A)
- (3) (A) is true but (R) is false
- (4) Both (A) and (R) are false

13. Assertion (A): The species is a group of interbreeding individuals reproductively isolated from the other species populations.

Reason (R): Prokaryotes cannot be kept under different species on the basis of reproductive isolation.

- (1) Both (A) & (R) are true and the (R) is the correct explanation of the (A)
- (2) Both (A) & (R) are true but the (R) is not the correct explanation of the (A)
- (3) (A) is true but (R) is false
- (4) Both (A) and (R) are false

14. Assertion (A): Most of the species are identified on the basis biological species concept.

Reason (R): Interbreeding can be practically used as routine criterion for identification.

- (1) Both (A) & (R) are true and the (R) is the correct explanation of the (A)
- (2) Both (A) & (R) are true but the (R) is not the correct explanation of the (A)
- (3) (A) is true but (R) is false
- (4) Both (A) and (R) are false

15. Assertion (A): Higher the taxonomic category, greater is the difficulty of determining the relationship to other taxa at the same level.

Reason (R): The higher taxonomic categories are more exclusive and the lower taxonomic categories are more inclusive in nature.

- (1) Both (A) & (R) are true and the (R) is the correct explanation of the (A)
- (2) Both (A) & (R) are true but the (R) is not the correct explanation of the (A)
- (3) (A) is true but (R) is false
- (4) Both (A) and (R) are false

16. What do A, B and C represent in the given scientific name?

Mangifera	Indica	Linn
A	B	C

- (1) Author's name, specific name and generic name
- (2) Specific name, generic name and author's name
- (3) Generic name, specific name and author's name
- (4) Generic name, author's name and specific name

17. Fill in the blanks and choose the correct option:-

	Category	Mango	Housefly	Man
(i)	Class	Dicotyledonae	A	Mammalia
(ii)	Order	B	Diptera	C
(iii)	Family	Anacardiaceae	D	Hominidae

	A	B	C	D
(1)	Insecta	Sapindales	Primata	Muscidae
(2)	Arthropoda	Polymoniales	Primata	Muscidae
(3)	Insecta	Sapindales	Mammalia	Canidae
(4)	Arthropoda	Poales	Mammalia	Felidae

18. Select the **incorrect** match:-

	Rank	Suffix
(1)	Division	- phyta
(2)	Family	- phyceae
(3)	Order	- ales
(4)	Class	- opsida

19. Match **column-I** with **column-II**.

	Column I		Column II
(a)	Phylum	(i)	Hominidae
(b)	Order	(ii)	<i>Homo</i>
(c)	Class	(iii)	Chordata
(d)	Genus	(iv)	Primata
(e)	Family	(v)	Mammalia

- (1) a-(ii), b-(iv), c-(v), d-(iii), e-(i)
- (2) a-(iii), b-(v), c-(ii), d-(i), e-(iv)
- (3) a-(i), b-(iii), c-(ii), d-(iv), e-(v)
- (4) a-(iii), b-(iv), c-(v), d-(ii), e-(i)

Exercise 4 (Previous Year's Questions)

1. Nomenclature is governed by certain universal rules. Which one of the following is contrary to the rules of nomenclature?
[NEET-I – 2016]
- (1) Biological names can be written in any language
 - (2) The first word in a biological name represents the genus name and second is specific epithet
 - (3) The names are written in latin and are italicized
 - (4) When written by hand, the names are to be underlined
2. Match **Column I** with **Column II** for housefly classification and select the correct option using the codes given below:
[NEET-I – 2016]
- | Column-I | Column-II |
|------------|-----------------|
| (a) Family | (i) Dipetera |
| (b) Order | (ii) Arthropoda |
| (c) Class | (iii) Muscidae |
| (d) Phylum | (iv) Insecta |
- Codes:**
- | (a) | (b) | (c) | (d) |
|-----------|-------|------|-------|
| (1) (iii) | (ii) | (iv) | (i) |
| (2) (iv) | (iii) | (ii) | (i) |
| (3) (iv) | (ii) | (i) | (iii) |
| (4) (iii) | (i) | (iv) | (ii) |
3. Study the four statements (A-D) given below and select the two correct ones out of them:
[NEET-2019]
- (A) Definition of biological species was given by Ernst Mayr
 - (B) Photoperiod does not affect reproduction in plant
 - (C) Binomial nomenclature system was given by R. H. Whittaker
 - (D) In unicellular organisms, reproduction is synonymous with growth
- The two correct statement are:
- | | |
|-------------|-------------|
| (1) C and D | (2) A and D |
| (3) A and B | (4) B and C |
4. Select the correctly written scientific name of Mango which was first described by Carolus Linnaeus:
[NEET-2019]
- (1) *Mangifera indica*
 - (2) *Mangifera Indica*
 - (3) *Mangifera indica* Car. Linn.
 - (4) *Mangifera indica* Linn.
5. Which of the following is against the rules of ICBN?
[NEET-2020]
- (1) Hand written scientific names should be underlined.
 - (2) Every species should have a generic name and a specific epithet.
 - (3) Scientific names are in Latin and should be italicized.
 - (4) Generic and specific names should be written starting with small letters.
6. Which one of the following belongs to the family muscidae?
[NEET-2021]
- | | |
|---------------|-----------------|
| (1) Fire fly | (2) Grasshopper |
| (3) Cockroach | (4) House fly |
7. In the taxonomic categories which hierarchical arrangement in ascending order is correct in case of animals?
[NEET-2022]
- (1) Kingdom ← Phylum ← Class ← Order ← Family ← Genus ← Species
 - (2) Kingdom ← Class ← Phylum ← Family ← Order ← Genus ← Species
 - (3) Kingdom ← Order ← Class ← Phylum ← Family ← Genus ← Species
 - (4) Kingdom ← Order ← Phylum ← Class ← Family ← Genus ← Species
8. 'X' and 'Y' are the components of Binomial nomenclature. This naming system was proposed by 'Z':
[NEET-2023] (MANIPUR)
- (1) X-Generic name, Y-Specific epithet, Z-Carolus Linnaeus
 - (2) X-Specific epithet, Y-Generic name, Z-R.H. Whittaker
 - (3) X-Specific epithet, Y-Generic name, Z-Carolus Linnaeus
 - (4) X-Generic name, Y-Specific epithet, Z-R.H. Whittaker

ANSWER KEYS

Exercise 1.1

Que.	1	2	3	4	5	6	7	8	9	10										
Ans.	4	2	1	3	4	2	4	4	2	2										

Exercise 1.2

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	2	1	2	3	4	1	4	3	3	4	1	4	1	4	1	4	3	1	3	3

Exercise 1.3

Que.	1	2	3	4	5	6	7													
Ans.	2	3	1	2	4	1	2													

Exercise 1.4

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	2	4	2	3	3	3	1	4	4	1	2	3	3	4	2	1	4	4	3	1
Que.	21																			
Ans.	1																			

Exercise 2

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	2	4	2	3	3	2	1	4	3	2	3	4	3	1	1	1	1	2	4	2

Exercise 3

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	
Ans.	1	1	4	1	1	3	1	4	1	3	2	3	2	1	1	3	1	2	4	

Exercise 4 (Previous Year's Questions)

Que.	1	2	3	4	5	6	7	8												
Ans.	1	4	2	4	4	4	1	1												